Hyperspectral Image Fusion and Multitemporal Image Fusion by Joint Sparsity

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***Abstract*— Different image fusion systems have been developed to deal with the massive amounts of image data for different applications, such as remote sensing, computer vision, and environment monitoring. However, the generalizability and versatility of these fusion systems remain unknown. This article proposes an efficient regularization framework to achieve different kinds of fusion tasks accounting for the spatiospectral and spatiotemporal variabilities of the fusion process. A joint minimization functional is developed by taking an advantage of a composite regularizer for enforcing joint sparsity in the gradient domain and the frame domain. The proposed composite regularizer is composed of the Hessian Schatten-norm regularization and contourlet-based regularization terms. The resulting problems are solved by the alternating direction method of multipliers (ADMM). The effectiveness of the proposed method is validated in a variety of image fusion experiments: 1) hyperspectral (HS) and panchromatic image fusion; 2) HS and multispectral image fusion; 3) multitemporal image fusion (MIF); and 4) multi-image deblurring. Results show promising performance compared with state-of-the-art fusion methods.**

***Index Terms*— Alternating direction method of multipliers, image fusion, regularization framework.**

I. Introduction

THE development of optical Earth observation satellites has created a new wave of sensing impacts to the remote sensing community [1], [2]. However, hyperspectral (HS) image suffers from low spatial resolution. Multispectral (MS) image has a lower spectral resolution and higher spatial resolution. Fortunately, image fusion is an effective way to yield highly informative data, which affects the potential applications, such as land use monitoring [3] and location of wildfires [4], [5].

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A large number of remote sensing image fusion methods have been developed [6], [7]. There are several types of image fusion methods, such as HS image fusion including MS/HS fusion [8], [9], panchromatic (PAN)/HS fusion [10], [11], and multitemporal image fusion (MIF) [7]. However, most of the existing methods have been developed independently. Little attention has been paid to study the relationships between them. Notably, the generalizability and versatility of these fusion systems remain unknown. Thus, a more general framework is required to represent the fusion process, including spatiotemporal fusion [12] and spatiospectral fusion [13]. Meanwhile, the design of the regularization method for a good fusion performance is still an open problem [6].

A popular regularization method for remote sensing image fusion is total variation [8] (TV) seminorm. This method is adapted to fuse while preserving sharp edges, but it will, sometimes, lead to the undesired staircase effect [14]. The wavelet transform has been used in MS/HS fusion [15]. The wavelet transform is due to its success in its ability to approximating signals containing pointwise singularities. However, this transform is not so effective at dealing with edges in images. Furthermore, these methods may be corrupted by varying acquisition conditions, such as cloud or cloud shadow.

A common workaround to mitigate these issues is to incorporate a joint regularization method to model the important properties of natural images. By taking advantage of second-order information of the image, the Hessian Schatten-norm regularization method can capture features structure and imposes the structured sparsity in the gradient domain [14]. Furthermore, the contourlet-based regularization method is more effective in approximating smooth contours in images [16]. It has an advantage in providing a sparse representation (SP) for 2-D piecewise smooth signals resembling images. These recent advances motivate us to investigate the joint sparsity of the spatiospectral and spatiotemporal variabilities of the fusion process across different domains.

A. Scope and Contributions

In this work, we present a general regularization framework to improve the generalizability and versatility of the image fusion systems. The key idea is to carefully construct a relationship model by promoting joint sparsity in the gradient domain and the frame domain uses an optimization method

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to generate the final fused image. The contributions can be summarized as follows.

1) A general regularization framework is proposed to model the spatiospectral and spatiotemporal variabilities of the fusion process efficiently. A composite regularization approach is established for modeling joint sparsity in the gradient domain and the frame domain, i.e., Hessian Schatten-norm regularization and contourlet-based regularization terms.
2) Our approach is quite general in the sense that it is able to handle a variety of image fusion problems, including hyperspectral image fusion (PAN/HS fusion and MS/HS fusion), MIF, and multi-image deblurring. Efficient alternating minimization methods are developed to take full advantage of the complementary structural information across different domains. Empirical performance evaluations establish the advantages of the proposed method over other state-of-the-art algorithms.

B. Related Work

1) HS Image Fusion: HS image fusion refers to the problem of inferring images with both high spatial and spectral resolutions. The classical HS image fusion methods include PAN/HS fusion and MS/HS fusion.

Integrating and analyzing data from HS and PAN images, also known as HS pansharpening [17], have attracted extensive research efforts. Some kinds of strategies are presented. The first one is component substitution, such as principal components analysis (PCA)-based methods [18], [19] and intensity-hue-saturation-based methods [20], [21]. The second strategy is the multiresolution analysis that integrates the spatial details in a multiresolution manner. Some typical examples include smoothing filter-based intensity modulation (SFIM) [22] and regression-based high-pass modulation approach [23]. The third one is the Bayesian inference that infers probabilistic parameters based on the fusion model [24]. The fourth one is the variational method [10], [25], [26] that extracts features in the gradient domain and performs well compared with other algorithms.

Significant research efforts have been expended to MS/HS fusion [13], [27]–[29]. These methods can be roughly classified into five types, i.e., compressive sensing [30]–[32] (CS), [29] SP, multiresolution analysis [33]–[35], nonnegative matrix factorization [27] (NMF), and variational methods [8], [10]. Zhu *et al*. [30]–[32] presented a joint sparsity-promoting framework using distributed CS theory. Wei *et al*. [29] proposed a novel fusion method based on dictionary learning and sparse coding. In [33], the 3-D-wavelet transform was used to MS/HS fusion with average and substitution rule. Yokoya *et al.* [27] proposed a coupled NMF-based method with a linear spectral mixture model. Simões *et al*. [8] developed a subspace-based spatial enhancement method using TV regularization term.

2) Multitemporal Image Fusion: The concept of MIS [12], [36]–[38] has found its way into many disciplines. In [36], the wavelet transform was used to integrate the information from multisource images. Huang and Song [12] proposed a method built on a dictionary of local patches to spatiotemporal reflectance fusion. The theory of compressed sensing was utilized to model spatiotemporal variability [37]. Xu *et al*. [38] presented a spatiotemporal fusion model with unmixing regularized strategy. Note that few studies investigate the relationship between the spatiospectral and spatiotemporal variabilities of the fusion process.

In addition, cloud removal is significantly needed for MIF. Recently, some deep learning-based methods for cloud detection [39] and cloud removal [40] have been developed, such as the deep convolutional neural network [41] and the extended super-resolution convolutional neural network [42].

3) Multi-Image Deblurring: Multi-image deblurring aims to recover the clear image using multiple blurry and/or noisy observations, which could be seen as an alternative way to spatiotemporal fusion. Cai *et al*. [43] presented a blind motion deblurring method by considering the sparsity prior to motion blur kernel and image alignment. Zhang *et al*. [44] designed a multi-image blind deblurring algorithm with a coupled adaptive sparse prior. Jeon *et al*. [45] proposed a multi-image motion deblurring method by introducing the concept of complementary sets of fluttering patterns.

Finally, it can be seen that most image fusion techniques have been constrained to relatively specialized solutions that are lack of versatility to deal with complex tasks [46]. Although the joint sparsity model was used to represent the latent subspace of HS image [47], few studies investigate the relationship between them. Therefore, we introduce a composite regularization method to investigate the joint sparsity or behavior account for the spatiotemporal and spatiospectral variabilities.

This article is organized as follows. In Section II, basic notations and the details of the proposed regularization framework are provided. A joint minimization functional with a composite regularizer is defined. Then, four real-world applications are considered. In Section III, a detailed solution with the alternating direction method of multipliers (ADMMs) is presented and discussed. In Section IV, experimental results are presented. Finally, we summarize the results in Section V.

II. General Regularization Framework and Applications

A. General Regularization Framework With Joint Sparsity

In this article, we propose a general regularization framework with joint sparsity, which defines a joint minimization functional with a composite regularizer. More specifically, it enforces joint sparsity across different domains. Many new models for different kinds of fusion tasks within the proposed framework are presented. In this article, vectors are denoted by bold lower case letters and matrices by upper case ones.

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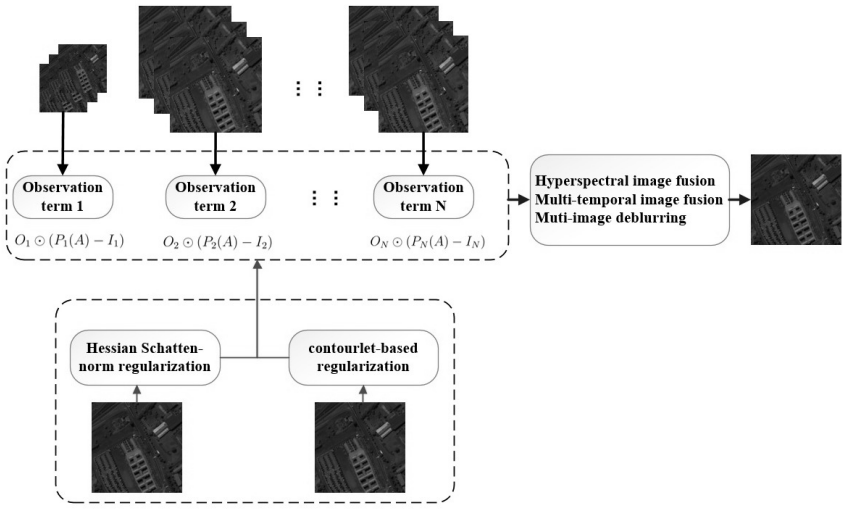


Fig. 1. General regularization framework with joint sparsity for HS image fusion, multitemporal image fusion, and multi-image deblurring.

The proposed regularization framework is presented as follows:

$$\min_{A\in B_{l,h}} \sum_{i}^{N} \frac{1}{2}\|O_i \odot (P_i(A) - I_i)\|_F^2 + \lambda_1 \sum_{j}^{D} HS(A_{(j,:)}) + \lambda_2 \sum_{j}^{D} \Phi(A_{(j,:)}) \quad (1)$$

where $A, I \in \mathbb{R}^{D\times K}$ represent the estimated image and the input multisource image, respectively. N is the number of multisource images. $K = m \times n$ denotes the number of pixels, and D represents total bands of A. m is the number of vertical pixels, and n is the number of horizontal pixels. $P_i(\cdot)$ is an operator representing some processes, such as blurring and subsampling. $A_{(j,:)}$ is the jth band of the estimated image. $HS(\cdot)$ denotes the Hessian Schatten-norm regularization term. $\Phi(\cdot)$ represents the frame representation-based regularization method with ℓ_1 norm, i.e., contourlet transformation [16]. O_i is assumed to be a componentwise matrix that denotes missing data (i.e., $O_i(u, v) = 0$ or 1), such as the location of cloud and cloud shadow. $B_{l,h}$ is the bounding constraint with $B_{l,h} = \{l \le A_{uv} \le h \forall u, v\}$. l and h are the lower and upper bounds, respectively. The structure of the general regularization framework is illustrated in Fig. 1.

The proposed composite regularizer aims to promote joint sparsity of the local structure of the domains, i.e., representing or extracting the local features across different domains. This approach models the joint sparsity on two complementary domains. First, the Hessian Schatten-norm regularization term is used to promote the sparsity on image gradient. Second, each band is assumed to be sparse after the application of contourlet transform [16]. It should be noted that the proposed regularization scheme is a convex composite functional.

B. Hessian Schatten-Norm Regularization

The Hessian Schatten-norm regularization term involves a second-order differential operator with the Schatten-norm [14]. The key idea of this regularization method is to capture curvature information related to the image pixel.

For simplicity, we use $x = A_{(j,:)}$ to represent the vector version of each band. This regularization term can be represented as follows:

$$HS(x) = \|\mathcal{H}x\|_{1,p} = \sum_{n=1}^{K} \|[\mathcal{H}x]_n\|_{\mathcal{S}_p} \quad \forall p \ge 1 \quad (2)$$

where $\mathcal{H}$ denotes the discrete Hessian operator. $\|\cdot\|_{1,p}$ denotes the mixed ℓ_1-$\mathcal{S}_p$ norm. $\|\cdot\|_{\mathcal{S}_p}$ represents the Schatten norm, which indicates ℓ_p norm on the singular-values $\sigma \in \mathbb{R}_+^{\min(m,n)}$ of a matrix. $[\mathcal{H}x]_n$ generates 2×2 symmetric matrix at the nth entry of x, where $n = jm + (i + 1)$ mapped by the coordinates (i, j).

It should be noted that $\|\mathcal{H}x\|_{1,p}$ has a strong relation to the sparsity-promoting group norms [14], [48]. It clusters features into groups and then enforces sparsity, which commonly appeared in the context of CS [48]. The Schatten norm of the Hessian matrix is to promote the low-rankness on $[\mathcal{H}x]_n$, which approximates the intensity variations in natural image. Moreover, the singular-value decomposition (SVD) in the Schatten norm for $[\mathcal{H}x]_n$ reduces to the spectral decomposition. Then, the eigenvalues of the Hessian at every coordinate correspond to the principal curvatures, which provides an efficient way to evaluate how the surface bends by different amounts in different directions at that point. Furthermore, this norm leads to piecewise linear reconstructions compared with the piecewise-constant reconstructions by TV regularization. Thus, the Hessian Schatten-norm regularization can avoid the staircase effect [14].

This operator provides the mapping: $\mathcal{H} : \mathbb{R}^K \to \mathcal{Q}$, where $\mathcal{Q} = \mathbb{R}^{K\times 2\times 2}$. For $\mathbf{x} \in \mathbb{R}^K$, $\mathcal{H}x$ is expressed as

$$[\mathcal{H}x]_n = \begin{bmatrix} \{\triangle_{r_1r_1}\, x\}_n & \{\triangle_{r_1r_2}\, x\}_n \\ \{\triangle_{r_2r_1}\, x\}_n & \{\triangle_{r_2r_2}\, x\}_n \end{bmatrix} \quad (3)$$

where $\triangle_{r_1r_1}$, $\triangle_{r_1r_2}$, and $\triangle_{r_2r_2}$ are the finite-difference operators that provide the discrete approximation of the second-order

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partial derivatives. $\{\cdot\}_n$ stands for the nth element of the argument, $n = 1, \ldots, K$. The standard representation of these operators on the image is

$$\{\triangle_{r_1 r_1}\, x\}_{i,j} = \begin{cases} x_{i+2,j} - 2x_{i+1,j} + x_{i,j}, & 1 \le i \le m-2 \\ x_{m-1,j} - x_{m,j}, & i \ge m-1 \end{cases} \quad (4)$$

$$\{\triangle_{r_1 r_2}\, x\}_{i,j} = \begin{cases} x_{i+1,j+1} - x_{i+1,j} - x_{i,j+1} + x_{i,j}, \\ \quad 1 \le i \le m-1 \text{ and } 1 \le j \le n-1 \\ 0, \quad \text{otherwise} \end{cases} \quad (5)$$

$$\{\triangle_{r_2 r_2}\, x\}_{i,j} = \begin{cases} x_{i,j+2} - 2x_{i,j+1} + x_{i,j}, & 1 \le j \le n-2 \\ x_{i,n-1} - x_{i,n}, & j \ge n-1. \end{cases} \quad (6)$$

C. Contourlet-Based Regularization

The contourlet-based regularization term is considered to encode directional structure in images, which provides the ability to capture the smoothness of the contour of images in different directions [16]. Assume that the observed scene can be represented by a linear combination of basis vectors $\{\phi_i\}_1^L$, and we have

$$x = \sum_{i=1}^{L} \langle x, \phi_i \rangle \phi_i \quad (7)$$

where $\langle \cdot, \cdot \rangle$ stands for the procedure of inner product, $\alpha_i = \langle x, \phi_i \rangle$ is used to denote the ith basis coefficient, and L denotes the number of bases. Let $\alpha = [\alpha_1, \ldots, \alpha_L]^{\mathrm{T}}$ and $\phi = [\phi_1, \ldots, \phi_L]$, and (7) can be rewritten as $x = \phi\alpha$. The problem of fusing local structural information can be formulated as an optimization problem to obtain a sparse vector $\alpha \in \mathbb{R}^L$. In this article, we consider the contourlet frame. It should be noted that $\phi\phi^T = I$. Thus, we have $x = \phi\alpha$ when $\alpha = \phi^T x$. Moreover, the contourlet-based regularization term can be expressed as

$$\Phi(x) \ = \|\phi^T x\|_1. \quad (8)$$

The contourlet-based regularization term with ℓ_1 norm is able to promote low-frequency information of images sparsely. In other words, the sparse coefficients are treated as the salient features of the image.

The choice of the contourlet basis is an interesting and complex topic. The contourlet transform is defined as a combination of Laplacian pyramid and direction filter bank [16]. We verified the filter structure experimentally. There are many kinds of orthogonal wavelets and biorthogonal wavelets, such as "Daubechies 9/7," "5/3," "Burt," "haar," and "pkvaN." After some experiments, we can see that the linear phase characteristics of biorthogonal wavelet are suitable for reducing the artifact. Thus, we set the pyramid filter as "9/7" and the directional filter "pkva."

D. Applications

Our framework obtains constructive objective function for the fusion process in various situations. It leads to some new models for different kinds of fusion tasks. We illustrate the flexibility of the proposed general regularization framework with the following four applications.

1) *HS and MS Image Fusion:*

$$\min_{A \in B_{l,h}} \frac{1}{2}\|I_h - EABM\| + \frac{\lambda_m}{2}\|I_m - REA\| + \lambda_1 \sum_{j=1}^{D} HS(A_{(j,:)}) + \lambda_2 \sum_{j=1}^{D} \Phi(A_{(j,:)}) \quad (9)$$

where I_h and I_m are the input HS and MS images, respectively, E denotes a matrix where the columns span the subspace as A. B stands for the point spread function by HS sensor, R denotes the spectral responses of the MS sensor, and M represents a downsampling operator on the image, which generates the lower resolution of the HS image. B and R can be calculated by the method in [8].

2) *HS and Panchromatic Image Fusion:*

$$\min_{A \in B_{l,h}} \frac{1}{2}\|I_h - EABM\| + \frac{\lambda_p}{2}\|I_p - REA\| + \lambda_1 \sum_{j=1}^{D} HS(A_{(j,:)}) + \lambda_2 \sum_{j=1}^{D} \Phi(A_{(j,:)}) \quad (10)$$

where I_h and I_p represent the input HS and panchromatic images, respectively. The meanings of B and R are the same thing in (9).

3) *Multitemporal Image Fusion:*

$$\min_{A \in B_{l,h}} \frac{1}{2} \sum_{i}^{N} \|O_i \odot (A - I_i)\|_F^2 + \lambda_1 \sum_{j=1}^{D} HS(A_{(j,:)}) + \lambda_2 \sum_{j=1}^{D} \Phi(A_{(j,:)}) \quad (11)$$

where O_i denotes the weighting matrix, I_i represents ith observed image of N total images, and P_i denotes the identity matrix. It is noted that not all pixels of the scene are available. Some methods are required to detect and select the corrupted pixels.

4) *Multi-Image Deblurring:*

$$\min_{A \in B_{l,h}} \frac{1}{2} \sum_{i}^{N} \|K_i A - I_i\|_F^2 + \lambda_1 \sum_{j=1}^{D} HS(A_{(j,:)}) + \lambda_2 \sum_{j=1}^{D} \Phi(A_{(j,:)}) \quad (12)$$

where I_i denotes the blurry image, $K_i = P_i$ represents the blur kernel, and N denotes the number of the input images. It is noted that O_i is a positive weighting matrix where the elements are all ones. For simplicity, we ignore the symbol O_i.

III. Alternating Minimization

In this section, we focus on the problem of MS/HS fusion. Particularly, we can also get the similar results for the other fusion tasks. The framework of ADMM is used to solve (9). The key procedure of ADMM is to determine the variables for splitting. Four auxiliary variables are introduced, i.e., U_1–U_4. For simplicity, the composite regularization term is

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reformulated as $HS(A) = \sum_{j=1}^{D} HS(A_{(j,:)})$ and $\Psi(\phi^T A) = \sum_{j=1}^{D} \Phi(A_{(j,:)})$, respectively.

Thus, the joint objective functional for MS/HS fusion can be represented by

$$\begin{aligned} \min\ & \frac{1}{2}\|I_h - EU_1M\| + \frac{\lambda_m}{2}\|I_m - REU_2\| \\ & + \lambda_1 HS(U_3) + \lambda_2 \Psi(U_4) \\ \text{s.t. } & U_1 = AB, U_2 = A, U_3 = A, U_4 = \phi^T A. \end{aligned} \tag{13}$$

The resulting augmented Lagrangian function $\mathcal{L}$ is expressed by

$$\begin{aligned} &\mathcal{L}(A, U_1, U_2, U_3, U_4, V_1, V_2, V_3, V_4) \\ &= \frac{1}{2}\|I_h - EU_1M\|_F^2 + \frac{\mu}{2}\|AB - U_1 - V_1\|_F^2 \\ &\quad + \frac{\lambda_m}{2}\|I_m - REU_2\|_F^2 + \frac{\mu}{2}\|A - U_2 - V_2\|_F^2 \\ &\quad + \lambda_1 HS(U_3) + \frac{\mu}{2}\|A - U_3 - V_3\|_F^2 \\ &\quad + \lambda_2 \Psi(U_4) + \frac{\mu}{2}\|\phi^T A - U_4 - V_4\|_F^2 \end{aligned} \tag{14}$$

where the vectors V_1, V_2, V_3, and V_4 are the Lagrange multipliers, and $\mu > 0$ is the half quadratic penalty parameter. We obtain the following sequence of subproblems, which can be represented by:

$$\begin{cases} A^{k+1} = \min_A \mathcal{L}\left(A, U_1^k, U_2^k, U_3^k, U_4^k, V_1^k, V_2^k, V_3^k, V_4^k\right) \\ U_1^{k+1} = \min_{U_1} \mathcal{L}\left(A^{k+1}, U_1, U_2^k, U_3^k, U_4^k, V_1^k, V_2^k, V_3^k, V_4^k\right) \\ U_2^{k+1} = \min_{U_2} \mathcal{L}\left(A^{k+1}, U_1^{k+1}, U_2, U_3^k, U_4^k, V_1^k, V_2^k, V_3^k, V_4^k\right) \\ U_3^{k+1} = \min_{U_3} \mathcal{L}\left(A^{k+1}, U_1^{k+1}, U_2^{k+1}, U_3, U_4^k, V_1^k, V_2^k, V_3^k, V_4^k\right) \\ U_4^{k+1} = \min_{U_4} \mathcal{L}\left(A^{k+1}, U_1^{k+1}, U_2^{k+1}, U_3^{k+1}, U_4, V_1^k, V_2^k, V_3^k, V_4^k\right) \end{cases} \tag{15}$$

and the corresponding Lagrange multipliers are updated as

$$\begin{aligned} V_1^{k+1} &= V_1^k - \left(A^{k+1}B - U_1^{k+1}\right) \\ V_2^{k+1} &= V_2^k - \left(A^{k+1} - U_2^{k+1}\right) \\ V_3^{k+1} &= V_3^k - \left(A^{k+1} - U_3^{k+1}\right) \\ V_4^{k+1} &= V_4^k - \left(\phi^T A^{k+1} - U_4^{k+1}\right). \end{aligned} \tag{16}$$

The overall procedure is summarized in Algorithm 1. A stopping criterion is used for the problems of image fusion in this article [49]. Thus, we ran the fusion method for 150 iterations in every case.

There are several subproblems arose from the solution.

1) *A-Subproblem:* The solution is obtained by dropping terms that do not involve A in (14). This yields

$$\begin{aligned} A^{k+1} = \min_A\ & \frac{\mu}{2}\left\|AB - U_1^k - V_1^k\right\|_F^2 \\ & + \frac{\mu}{2}\left\|A - U_2^k - V_2^k\right\|_F^2 + \frac{\mu}{2}\left\|A - U_3^k - V_3^k\right\|_F^2 \\ & + \frac{\mu}{2}\left\|\phi^T A - U_4^k - V_4^k\right\|_F^2. \end{aligned} \tag{17}$$

This problem can be solved after considering the first-order optimality condition, which can be expressed as follows:

$$\begin{aligned} A^{k+1} = [BB^T + I + I + \phi\phi^T]^{-1}\big[&\left(U_1^k + V_1^k\right)B^T \\ &+ \left(U_2^k + V_2^k\right) + \left(U_3^k + V_3^k\right) + \left(U_4^k + V_4^k\right)\big]. \end{aligned} \tag{18}$$

Due to $\phi\phi^T = I \forall \phi$, thus, (18) can be rewritten as follows:

$$\begin{aligned} A^{k+1} = \left[BB^T + I + I + I\right]^{-1}\big[&\left(U_1^k + V_1^k\right)B^T \\ &+ \left(U_2^k + V_2^k\right) + \left(U_3^k + V_3^k\right) + \left(U_4^k + V_4^k\right)\big]. \end{aligned} \tag{19}$$

where I denotes the identity matrix.

2) *U_1-Subproblem:* The U_1-subproblem is given by

$$U_1^{k+1} = \min_{U_1} \frac{1}{2}\|I_h - EU_1M\|_F^2 + \frac{\mu}{2}\left\|A^{k+1} - U_1 - V_1^k\right\|_F^2. \tag{20}$$

The closed-form solution of (20) can be obtained by

$$\begin{aligned} U_1^{k+1}M = &[E^T E + \mu I]^{-1} \\ &\times \left[E^T I_h + \mu\left(A^{k+1}B - V_1^k\right)\right]M. \end{aligned} \tag{21}$$

3) *U_2-Subproblem:* The U_2 minimization problem is given by

$$U_2 = \min_{U_2} \frac{1}{2}\|I_m - REU_2\|_F^2 + \frac{\mu}{2}\|A^{k+1} - U_2 - V_2^k\|_F^2. \tag{22}$$

A closed-form solution to (22) is provided as follows:

$$\begin{aligned} U_2^{k+1} = &[\lambda_m E^T R^T RE + \mu I]^{-1} \\ &\times \left[\lambda_m E^T R^T I_m + \mu\left(A^{k+1} - V_2^k\right)\right]. \end{aligned} \tag{23}$$

4) *U_3-Subproblem:* The U_3-subproblem is given by

$$U_3^{k+1} = \min_{U_3} \frac{\mu}{2}\left\|A^{k+1} - U_3 - V_3^k\right\|_F^2 + \lambda_1 HS(U_3). \tag{24}$$

For simplicity, we use $U_0 = A^{k+1} - V_3^k$. Then, (24) can be rewritten as

$$U_3^{k+1} = \min_{U_3} \frac{\mu}{2}\|U_0 - U_3\|_F^2 + \lambda_1 HS(U_3). \tag{25}$$

Algorithm 1 Alternating Minimization for MS/HS Fusion

Require: HS image I_h, MS image I_m, regularization parameters λ_1, λ_2, matrices R, B, and E, $k = 0$, the maximum number of iterations $N \geq 0$.

for k=0:N **do**

Update A^{k+1} by Eq. (19)
Update U_1^{k+1} by Eq. (21)
Update U_2^{k+1} by Eq. (23)
Update U_3^{k+1} by Eq. (27)
Update U_4^{k+1} by Eq. (29)
$V_1^{k+1} = V_1^k - (A^{k+1}B - U_1^{k+1})$
$V_2^{k+1} = V_2^k - (A^{k+1} - U_2^{k+1})$
$V_3^{k+1} = V_3^k - (A^{k+1} - U_3^{k+1})$
$V_4^{k+1} = V_4^k - (\phi^T A^{k+1} - U_4^{k+1})$

end for

Output: A

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It is noted that $HS(U_3)$ is the mixed ℓ_1-S_p norm. To perform U_3-update of (25), we need to do columnwise with the numerical solution in [14]. The resulting subfunction with the S_p norm multiplied by a constant is defined by

$$\text{prox}_{\frac{\lambda_1}{\mu}\|\cdot\|_{S_p}}(U_{0,:c}) = \min_{U_{3,:c}} \frac{1}{2}\|U_{3,:c} - U_{0,:c}\|_F^2 + \frac{\lambda_1}{\mu}\|U_{3,:c}\|_{S_p} \tag{26}$$

where $(\cdot)_{:c}$ denotes the cth column of a matrix. This operation is the proximal map of the Schatten S_p matrix-norm with order p. A closed-form of this proximal map with $p = 1$ is computed by

$$\text{prox}_{\frac{\lambda_1}{\mu}\|\cdot\|_{S_1}}(U_{0,:c}) = U\text{diag}(ST_{\frac{\lambda_1}{\mu}}(\sigma))V^H \tag{27}$$

where $ST_{\lambda_1/\mu}(\sigma) = \max(\sigma - (\lambda_1/\mu), 0)$ represents the soft-thresholding operator [50]. The SVD of the rearranged $U_{0,:c}$ is $U\Sigma V^H$. σ is the diagonal entries of Σ, which denotes the vector of singular values.

5) U_4-*Subproblem:* The U_4-subproblem is expressed as

$$U_4^{k+1} = \min_{U_4} \frac{\mu}{2}\left\|\phi^T A^{k+1} - U_4 - V_4^k\right\|_F^2 + \lambda_2\Psi(U_4). \tag{28}$$

The resulting solution is a columnwise soft-threshold function [51], that is

$$U_4^{k+1} = \max(|\alpha - \frac{V_4}{\mu}| - \frac{\lambda_2}{\mu}, 0) \cdot \text{sign}\left(\alpha - \frac{V_4}{\mu}\right) \tag{29}$$

where $\alpha = \phi^T A$.

The first term of (19) in the right-hand side, i.e., $[BB^T + I + I + I]^{-1}$, can be calculated by fast Fourier transform (FFT). Similarly, the constant matrices $[E^T E + \mu I]^{-1}$ and $E^T I_h$ in (21) can be precomputed by FFT. The computational complexities of these expressions are dominated by FFT. We follow the stopping criterion in [49] to terminate the iteration.

In this article, I_h has to be preprocessed. The data can be handled by projecting I_h onto a subspace with the selected dimension by truncated SVD. When the ground-truth images are available, these images are projected onto this subspace. The matrix E is estimated by vertex component analysis [52] (VCA), and the corresponding implementation can be obtained online.[1]

We provide a brief description of the solutions to (10)–(12). Three problems are summarized as follows.

1) Multi-image deblurring in (12)

$$\begin{aligned}\min_A \ & \frac{1}{2}\sum_i^N \|K_i A - I_i\|_F^2 + \lambda_1 HS(U_1) \\ & + \lambda_2\Psi(U_2) \\ \text{s.t. } & U_1 = A, U_2 = \phi^T A.\end{aligned} \tag{30}$$

Thus, the final objective functional can be provided as follows:

$$\begin{aligned}&\mathcal{L}(A, U_1, U_2, V_1, V_2) \\ &= \frac{1}{2}\sum_i^N \|K_i A_i - I_i\|_F^2 \\ &\quad + \lambda_1 HS(U_1) + \frac{\mu}{2}\|A - U_1 - V_1\|_F^2 \\ &\quad + \lambda_2\Psi(U_2) + \frac{\mu}{2}\|\phi^T A - U_2 - V_2\|_F^2.\end{aligned} \tag{31}$$

2) HS and panchromatic image fusion in (10)

$$\begin{aligned}\min_A \ & \frac{1}{2}\|I_h - EU_1 M\| + \frac{\lambda_p}{2}\|I_p - REU_2\| \\ & + \lambda_1 HS(U_3) + \lambda_2\Psi(U_4) \\ \text{s.t. } & U_1 = AB, U_2 = A, U_3 = A, U_4 = \phi^T A\end{aligned} \tag{32}$$

where I_p denotes the input panchromatic image. The final objective functional can be provided as follows:

$$\begin{aligned}&\mathcal{L}(A, U_1, U_2, U_3, U_4, V_1, V_2, V_3, V_4) \\ &= \frac{1}{2}\|I_h - EU_1 M\|_F^2 + \frac{\mu}{2}\|AB - U_1 - V_1\|_F^2 \\ &\quad + \frac{\lambda_p}{2}\|I_p - REU_2\|_F^2 + \frac{\mu}{2}\|A - U_2 - V_2\|_F^2 \\ &\quad + \lambda_1 HS(U_3) + \frac{\mu}{2}\|A - U_3 - V_3\|_F^2 \\ &\quad + \lambda_2\Psi(U_4) + \frac{\mu}{2}\|\phi^T A - U_4 - V_4\|_F^2\end{aligned} \tag{33}$$

where $\lambda_p > 0$ is a positive regularization parameter.

3) MIF in (11)

$$\begin{aligned}\min_{A\in B_{l,h}} \ & \frac{1}{2}\sum_i^t \|O_i(A - I_i)\|_F^2 + \lambda_1 HS(U_1) \\ & + \lambda_2\Psi(U_2) \\ \text{s.t. } & U_1 = A, U_2 = \phi^T A\end{aligned} \tag{34}$$

where O_i denotes the elementwise matrix by the method of cloud and cloud shadow removal [53]. Similarly, the final objective functional is provided by

$$\begin{aligned}&\mathcal{L}(A, U_1, U_2, V_1, V_2) \\ &= \frac{1}{2}\sum_i^t \|O_i(A - I_i)\|_F^2 \\ &\quad + \lambda_1\mathcal{H}(U_1) + \frac{\mu}{2}\|A - U_1 - V_1\|_F^2 \\ &\quad + \lambda_2\|U_2\|_1 + \frac{\mu}{2}\|\phi^T A - U_2 - V_2\|_F^2.\end{aligned} \tag{35}$$

IV. Performance Evaluation

A. Data Sets

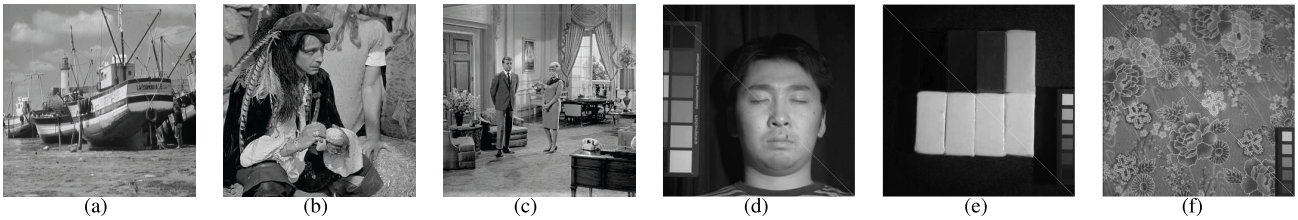
Several public data sets are evaluated here, i.e., Moffett field, Washington DC Mall,[2] Indian Pines [54], Kennedy Space Center [54], and multitemporal images.[3] The Moffett field data set is obtained by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor in California, USA, in 1997.

[1] http://www.lx.it.pt/ bioucas/code/demo_vca.zip

[2] https://engineering.purdue.edu/ biehl/MultiSpec/hyperspectral.html

[3] http://earthexplorer.usgs.gov/

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(a) (b) (c) (d) (e) (f)

Fig. 2. Illustration of the tested images for multi-image deblurring. (a) Boat. (b) Man. (c) Couple. (d) Face. (e) Clay. (f) Cloth.

TABLE I
NUMERICAL RESULTS OF MULTI-IMAGE DEBLURRING FROM THE IMAGES

		PSNR(dB)			SSIM(%)		
Images	Blur kernel	GAPG	Our($N=1$)	Our($N=6$)	GAPG	Our($N=1$)	Our($N=6$)
	Average	24.95	25.15	**27.29**	82.89	83.41	**88.64**
boat	Gaussian	25.18	25.37	**27.21**	83.99	84.48	**88.78**
	Starck-Murtagh	29.66	29.91	**31.66**	96.63	97.05	**98.39**
	Average	23.40	23.54	**25.26**	82.45	82.97	**88.54**
man	Gaussian	23.62	23.76	**25.18**	83.55	84.04	**88.66**
	Starck-Murtagh	27.71	27.93	**29.96**	95.91	96.40	**98.21**
	Average	24.65	24.79	**26.80**	81.27	81.81	**87.78**
couple	Gaussian	24.82	24.88	**26.65**	82.24	82.57	**87.52**
	Starck-Murtagh	29.57	29.83	**31.81**	95.99	96.45	**98.18**
	Average	31.71	31.78	**33.42**	96.84	96.90	**97.59**
face	Gaussian	31.82	31.88	**33.20**	97.12	97.18	**97.73**
	Starck-Murtagh	36.86	36.85	**38.79**	99.30	99.43	**99.63**
	Average	31.45	31.51	**32.83**	95.03	95.14	**95.65**
clay	Gaussian	31.71	31.74	**32.89**	95.31	95.43	**95.86**
	Starck-Murtagh	36.05	36.10	**37.49**	98.16	98.48	**98.78**
	Average	31.42	31.47	**32.60**	92.17	92.39	**93.85**
pompoms	Gaussian	31.62	31.66	**32.67**	92.67	92.90	**94.08**
	Starck-Murtagh	35.52	35.64	**36.95**	98.00	98.30	**98.80**
	Average	27.93	28.04	**29.70**	88.44	88.77	**92.00**
MEAN	Gaussian	28.12	28.21	**29.63**	89.14	89.43	**92.11**
	Starck-Murtagh	32.56	32.71	**34.44**	97.33	97.68	**98.66**

Its dimension is $480 \times 320 \times 176$ after removing the noisy bands. The Washington DC Mall data set was collected by the airborne-mounted Hyperspectral Digital Imagery Collection Experiment (HYDICE). Its size is $400 \times 300 \times 191$. The Indian Pines data set was recorded by the Reflective Optics System Imaging Spectrometer [55] (ROSIS) sensor.[4] The pixels of this data set are $400 \times 400 \times 200$. The Kennedy Space Center data set was collected by the airborne-mounted HYDICE. The resolution of each image of this data set is 500×400. Its bands are 176. The multitemporal images are obtained from EarthExplorer,[5] which may be corrupted by cloud occlusion of the Earth's surface and the shadows.

We perform the experiments in two different ways. First, four multiband images are used according to Wald's protocol [17]. Similarly, the original one is used as the reference image. The panchromatic or MS images are produced with three blur kernels, i.e., Gaussian with 3×3, $\sigma^2 = 0.8$, Starck–Murtagh filter with 5×5, and average with 7×7. Then, the resulting images are downgraded by the spectral responses of the Landsat 8 MS sensor. Finally, the zero-mean Gaussian noise is added to the images. In the second experiment, the masks for selecting the cloud and the corresponding shadow are created from the input multitemporal images. This procedure is implemented by the cloud removal algorithm [53] for detecting the cloud and shadow in each image. Finally, the produced images are adjusted by a guided filter [56]. These tests are conducted on a laptop equipped with 64-bit Windows 10 and performed on MATLAB 2014b.

B. Multi-Image Deblurring

We consider the applicability of the proposed approach to multi-image deblurring. Some numerical experiments under different blurring settings are carried out. The blur kernels are average, Gaussian, and Starck–Murtagh blur filters. The noise level is 1×10^{-3}. Some samples [57], including MS

[4]http://www.opairs.aero/rosis_en.html
[5]https://earthexplorer.usgs.gov/

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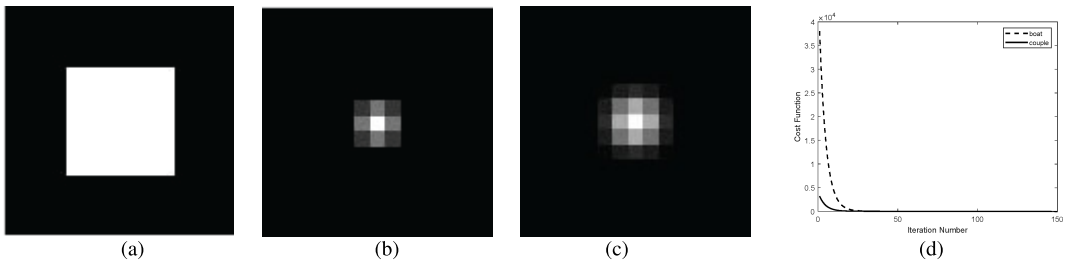


Fig. 3. Illustrative examples of the blur filters and the optimization process with 150 iterations. (a) Average filter (7×7). (b) Gaussian filter (3×3, 0.8). (c) Starck–Murtagh filter (5×5). (d) Evolution of the cost function during the optimization process for boat and couple. (a) Average filter. (b) Gaussian filter. (c) Starck–Murtagh filter. (d) Evolution of the cost function.

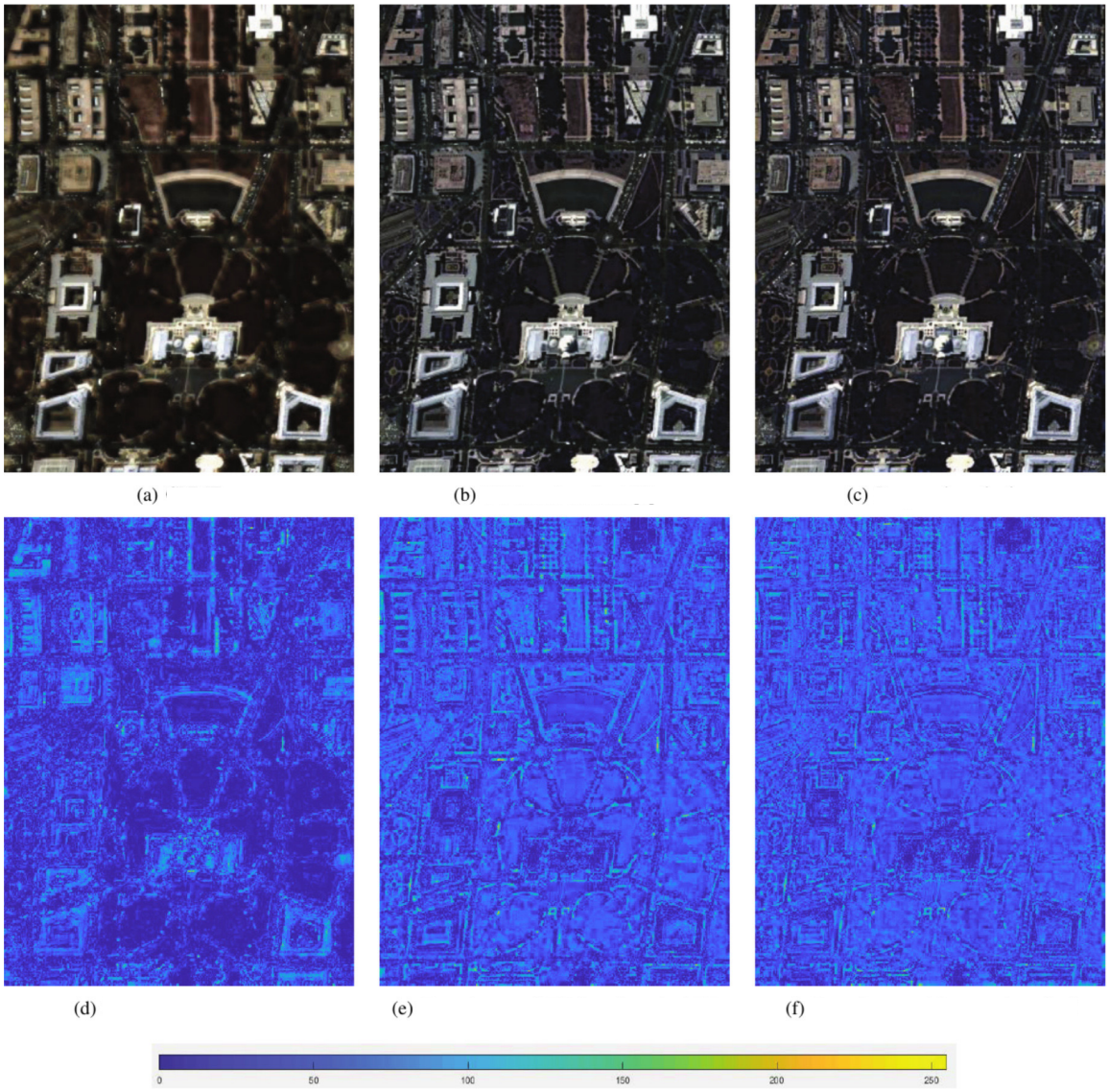


Fig. 4. Visual results and the corresponding error images of the test methods for the Washington DC Mall data set. Color image composites of the bands (30, 24, and 5). (a) By CNMF. (b) By TV. (c) By the proposed method. (d) Error image of CNMF. (e) Error image of the TV. (f) Error image of the proposed method. The blur kernel is Gaussian.

images and natural images,[6] are used, which is displayed in Fig. 2. The examples of three blur filters are presented in Fig. 3(a)–(c), respectively. The evaluation of cost function during the optimization process with 150 iterations for boat and couple is illustrated in Fig. 3(d). The regularization parameters of the referred methods are 1×10^{-3}.

The experiment results of multi-image deblurring are presented in Table I. Two performance measures are used,

[6]http://sipi.usc.edu/database/database.php?volume=misc

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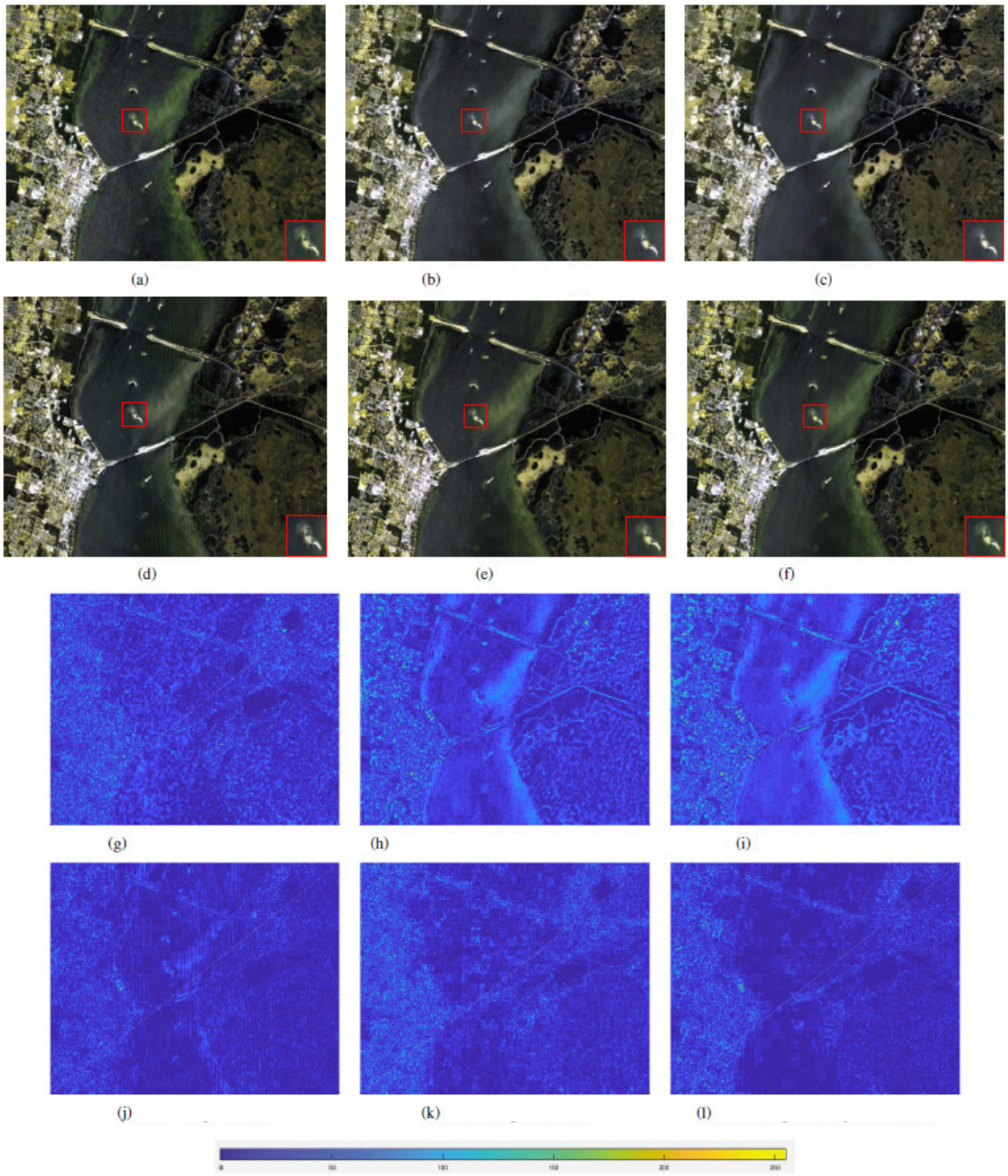


Fig. 5. Visual results and the corresponding error images of the test methods for the Kennedy Space Center data set. Color image composites of the bands (30, 24, and 5). (a) By CNMF. (b) By TV. (c) IFSU. (d) By CSU. (e) By LTRR. (f) By the proposed method. (g) Error image of CNMF. (h) Error image of TV. (i) Error image of IFSU. (j) Error image of CSU. (k) Error image of LTRR. (l) Error image of the proposed method.

TABLE II
NUMERICAL RESULTS OF HS PANSHARPENING BY THE MOFFETT FIELD DATA SET

Blur kernel	Gaussian			Starck-Murtagh			Average		
	ERGAS	SAM	UIQI(%)	ERGAS	SAM	UIQI(%)	ERGAS	SAM	UIQI(%)
CNMF [27]	2.1725	1.5695	91.30	2.3416	1.5205	90.87	2.4705	1.7422	88.07
TV [8]	1.3899	1.4777	92.62	1.3614	1.4818	92.71	1.4425	1.5938	92.46
Our	**1.0148**	**1.1255**	**94.96**	**0.9681**	**1.1383**	**94.86**	**1.0640**	**1.2605**	**94.17**

i.e., peak-signal-to-noise ratio (PSNR) and structure similarity (SSIM) index [58]. It can be seen that the proposed method is better than the TV-based method and the generalized accelerated proximal gradient [59] (GAPG) method. N denotes

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TABLE III
NUMERICAL RESULTS OF HS PANSHARPENING BY THE WASHINGTON DC MALL DATA SET

Blur kernel	Gaussian			Starck-Murtagh			Average		
	ERGAS	SAM	UIQI(%)	ERGAS	SAM	UIQI(%)	ERGAS	SAM	UIQI(%)
CNMF [27]	2.6802	1.7283	91.99	2.5011	1.6210	92.79	2.7278	1.5212	93.24
TV [8]	1.8031	0.9543	93.69	1.7722	1.0003	93.72	2.0783	1.0691	92.07
Our	**1.6182**	**0.6980**	**94.79**	**1.6686**	**0.7971**	**95.15**	**1.7999**	**0.9560**	**93.66**

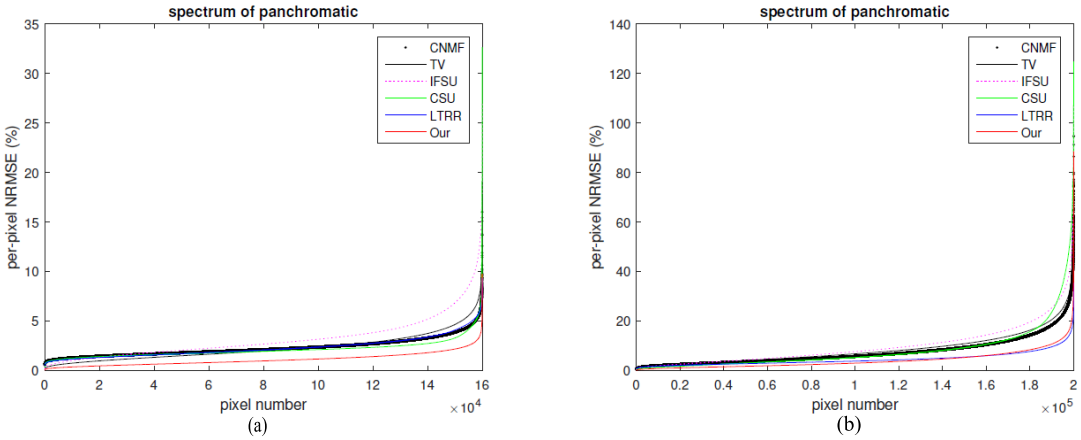


Fig. 6. Per-pixel NRMSE of various HS and MS fusion methods by the Indian Pines and Washington DC Mall data sets in ascending order. (a) For spectrum of panchromatic image of the Indian Pines data set. (b) For spectrum of panchromatic image of the Washington DC Mall data set.

TABLE IV
NUMERICAL RESULTS OF HS AND MS IMAGE FUSION BY THE INDIAN PINES DATA SET

Blur kernel	Gaussian			Starck-Murtagh			Average		
	ERGAS	SAM	UIQI(%)	ERGAS	SAM	UIQI(%)	ERGAS	SAM	UIQI(%)
CNMF [27]	1.1411	2.1211	88.11	1.1020	2.0935	88.67	1.0824	2.0946	88.15
TV [8]	1.1223	0.9646	89.31	1.1111	0.9697	89.79	1.1418	0.9905	89.41
IFSU [9]	1.1017	0.9568	90.02	1.0856	0.9468	90.17	1.1374	0.9871	89.57
CSU [62]	0.8291	1.6270	88.49	0.8360	1.6754	87.82	0.9828	1.6508	86.45
LTRR [63]	0.8681	1.7492	92.45	0.9378	1.9313	91.33	0.8829	1.8186	92.26
Our	**0.5570**	**0.4549**	**96.13**	**0.5005**	**0.4523**	**96.64**	**0.4810**	**0.4435**	**96.90**

the input of the noisy image. Moreover, the inner iteration numbers of the GAPG method is 100. It can be seen that the proposed method has the best performances for all testing images. The symbol "MEAN" denotes the average values of all the numerical results.

C. Panchromatic and HS Image Fusion

Next, we consider the fusion of HS and panchromatic images. Three widely used quality measures are considered, i.e., error relative global dimensional de synthèse [60] (ERGAS), spectral angle mapper (SAM), and universal image quality index [61] (UIQI). The numerical results of these data sets are illustrated in Tables II and III. It should be noted that the Gaussian white noise is added to the HS image (SNR = 30 dB) and the panchromatic image (SNR = 40 dB). The proposed method is found to outperform other fusion methods in terms of ERGAS, SAM, and UIQI values. The visual results of these fusion methods are displayed in Fig. 4. The fusion result of the proposed method is presented in Fig. 4(a). Fig. 5(d)–(f) shows the error images using competing approaches from the Washington DC Mall data set. We observe that the proposed method outperforms other fusion methods. It is worth noting that the proposed method can restore an edge-enhancing one, which is consistent with the improvements in the quality measures.

For quantitative comparison, we use the similar regularization parameters on each data set. In the first experiment on the Moffett Field data set, the regularization parameters λ_p, λ_1, and λ_2 are chosen to be 5×10^{-3}, 5×10^{-3}, and 1×10^{-4}, respectively. For the second experiment on the Washington DC Mall data set, the regularization parameters are chosen to be $\lambda_p = 2 \times 10^{-4}$, $\lambda_1 = 2 \times 10^{-4}$, and $\lambda_2 = 1 \times 10^{-4}$.

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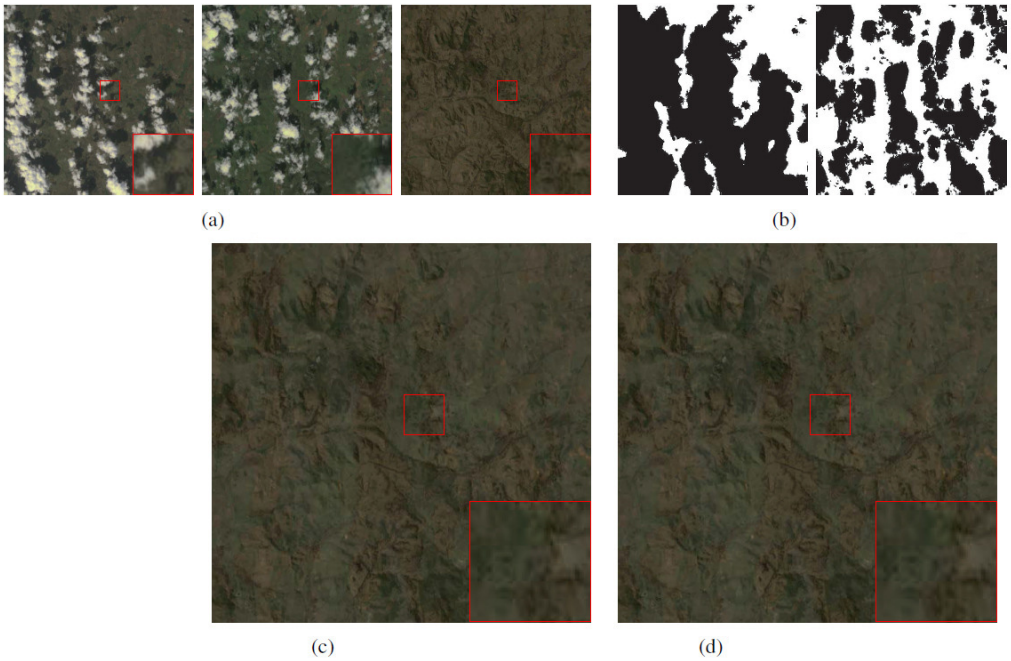


Fig. 7. Visual results of multitemporal image fusion. (a) Input three images. (b) Masks of three input images. (c) Proposed method. (d) TV-based method [8].

TABLE V
NUMERICAL RESULTS OF HS AND MS IMAGE FUSION BY THE KENNEDY SPACE CENTER DATA SET

Blur kernel	Gaussian			Starck-Murtagh			Average		
	ERGAS	SAM	UIQI(%)	ERGAS	SAM	UIQI(%)	ERGAS	SAM	UIQI(%)
CNMF [27]	2.5084	3.5401	88.35	2.0503	3.2695	90.10	2.5663	3.1558	90.81
TV [8]	2.3365	2.9337	92.35	2.3445	3.0493	92.30	1.8838	2.9798	92.59
IFSU [9]	2.7817	3.1066	90.99	2.8002	3.2134	91.03	2.9024	3.5947	90.35
CSU [62]	4.1147	4.0339	88.74	4.6138	4.3552	87.46	4.5894	4.1818	88.98
LTRR [63]	1.5520	3.2969	93.32	1.5997	3.4963	92.41	1.6633	3.5516	92.63
Our	**1.4744**	**2.1578**	**97.30**	**1.4348**	**2.1949**	**97.24**	**1.4348**	**2.1949**	**97.24**

D. HS and MS Image Fusion

The proposed method is applied to fuse HS and MS images. We compare the proposed method with five state-of-the-art methods, i.e., the TV-based method [8], multiband image fusion with spectral unmixing [9] (IFSU), coupled nonnegative matrix factorization [27] (CNMF)-based method, coupled spectral unmixing-based method [62] (CSU), and low tensor-train rank representation (LTRR)-based method [63]. In our experiment, we set $\lambda = 1 \times 10^{-3}$ for TV-based method. Besides, default parameters of the IFSU, CNMF, and CSU are used. For the LTRR, the number of clusters is set as $k = 160$, and the regularization parameter is set as $\lambda = 1 \times 10^{-2}$.

The numerical results of HS and MS image fusion are reported in Tables IV and V. The best result in each table is in bold font. It can be seen that the proposed method outperforms other fusion methods.

For visual comparisons, we provide the fusion results in Fig. 5(a)–(f). The selected part is enlarged and displayed in the bottom right corner. In our experiments, the blur kernel for downsampling is Gaussian with 3×3, and $\sigma^2 = 0.8$. Considering the results in Fig. 5(l), the proposed method can reconstruct local fine information. Fig. 5(g)–(l) depicts the error images of the test methods for the Kennedy Space Center data set, which helps to understand the differences of the test methods. As we can see from the error images, the fusion results produced by LTRR and the proposed method have fewer errors than those produced by the methods of CNMF, TV, IFSU, and CSU.

Fig. 6 shows the plots of the sorted per-pixel normalized root mean-square error (NRMSE) values of the referred fusion methods. NRMSE per-pixel is defined as

$$\frac{\|x_k - \hat{x}_k\|^2}{\|x_k\|^2}$$

where x_k and $\hat{x}_k$ denote the column of the image X and the corresponding fused image $\hat{x}$. The performance advantage of our method is found to outperform the other fusion methods over the spectrum of panchromatic image.

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E. Multitemporal Image Fusion

In this section, we apply our method to real multitemporal data and investigate the effects of various imaging conditions. However, some procedures are carried out. We first preprocessed the data to detect cloud and cloud shadow. There are two steps to detect the cloud and cloud shadow. First, a preidentification method described in [53] is utilized to locate the cloud or cloud shadow roughly. Then, a guided filter [56] is used to correct the results. In this experiment, the regularization parameters are $\lambda_1 = \lambda_2 = 1 \times 10^{-4}$.

The fused results of the test methods are demonstrated in Fig. 7. The samples of the data set are provided in Fig. 7(a), respectively. The masks for the cloud and cloud shadow are presented in Fig. 7(b), which indicates the locations of cloud and cloud shadow. The pictorial result of the proposed method is provided in Fig. 7(c). The selected part is enlarged and displayed in the bottom right corner. As we can see from the figure, the proposed framework can integrate more spatiotemporal information.

V. Conclusion

In this article, we propose an efficient regularization framework for HS image fusion and MIF. A joint minimization functional with a composite regularizer is defined. This regularizer considers the joint sparsity of the local structures in the gradient domain and the frame domain, which is a hybrid regularization mechanism with the Hessian Schatten-norm regularization and the contourlet-based regularization. The results demonstrate that the proposed method facilitates a high-quality image fusion. Numerical experiments on both simulated and real image data demonstrate that the proposed method provides an alternative way to represent the spatiospectral and spatiotemporal variabilities of the fusion process.

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